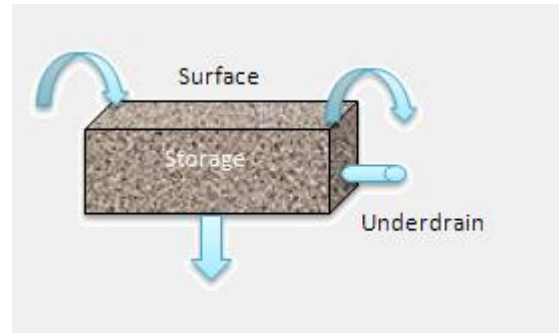


SWMM Modeling Applications

1.5 Infiltration Trench

You work for XYZ Company and have just been asked to substitute an infiltration basin for the bioretention from Example 1.3.

This exercise will help you understand how to create models quickly from existing models, how to adjust time series and other data when using a design storm as opposed to a continuous simulation, and will assess two approaches for modeling infiltration trenches. The first uses the classical infiltration trench as an LID practice. The second example uses a hybrid approach to simulate infiltration with storage on top, which is a similar to the function of many underground infiltration units.



You will create these models from the bioretention (long-term simulation) model on the USB.

Model 1 Directions:

1. Copy the directory \Bioretention\ but rename it to \InfiltrationBasin\
2. Open "Bioretention_LongTermSim.inp" in EPASWMM
3. Immediately save it as "Infiltration4_longterm.inp"
4. Rename the LID subcatchment to "infil".
5. Modify the outlet of subcatchment LOT to be infil.
6. Open LID control (under Hydrology) and change LID Type to Infiltration trench.
7. Under Storage, change thickness to 24 inches, and under Drain, change Offset height to 6 inches, everything else remains the same in the LID
8. Open Subcatchment Infil, and the LID editor, and make sure "LID occupies 100% of subcatchment" box is checked.
9. Run the simulation for 10 years, 1990-2000.

Model 2 directions:

1. For the above file, Save it again with a new name, "Infiltration4.inp"
2. Open Subcatchment Infil, and the LID editor, and make sure "LID occupies 100% of subcatchment" box is checked.
3. Load Chicago_10yr time series.
4. Change simulation date/times to start 01/01/2015, 0:00, end 1/3/2015 0:00.
5. Change time series in Rainage to Chicago10yr.
6. Check the weir, orifice, storage node, new outfall, new storage node, and new storage curve match the values in the tables provided.
7. Run the simulation.

Model 3 directions:

1. Save the model again with a new name, “Infiltration5.inp”
2. Rename outfall “Out2”
3. Make a new outfall to the North (above) and slightly east (right) of the existing outfall.
4. Rename this outfall, “Out1”.
5. Delete the only junction, note it automatically deletes the connections.
6. Insert a new Storage node.
7. Rename it “ST1”.
8. Insert a new orifice link from ST1 to Out1.
9. Insert a new weir link between the ST1 and Out2.
10. Connect subcatchment “Lot” to ST1.
11. Modify the outlet of subcatchment LOT to be ST1.
12. Connect subcatchment “Infil” to Out2.
13. Modify the outlet of subcatchment Infil to be Out2.
14. Create a new storage curve, “StorageCurve1”.
15. Fill in/correct attribute tables for the weir, the orifice, the storage node, the new outfall (Out1), and storage curve to match the tables provided.
16. Run the simulation for 10-year storm.
17. Load Chicago_10yr time series.
18. Change simulation date/times to start 01/01/2015, 0:00, end 1/2/2015 0:00.
19. Change time series in Raingage to Chicago_10yr

Model 4 directions:

1. For the model, “Infiltration5.inp”, save it with a new name, “Infiltration5_longterm.inp”
2. Delete Chicago_10yr time series, add ChicagoCont (and insert external data file, “Chicago_Rainfall.dat”).
3. Change raingage to use time series “ChicagoCont”
4. Change date/time of simulations to start on 1/1/1990 and end on 1/1/2000.
5. Run simulation

Please answer the following questions:

- 1) What are the differences in peak flow at the outlet, in a 10-yr storm between infiltration4 and infiltration5 models? What do you think accounts for the differences in peak?
- 2) What are the total infiltration and groundwater levels for the long term simulation, infiltration4.long term?

Table 1: Weir Characteristics

Weir	
Name	2
Type	Transvers
Height	1
Length	1
Side Slope	.5
Inlet offset	2
Discharge Coefficient	3.33
Flap Gate	No
Can Surge	Yes

Table 2: Orifice Characteristics

Orifice	
Name	1
Type	Bottom
Shape	Circular
Height	1
Width	0
Inlet offset	0
Discharge Coefficient	0.65
Flap Gate	No
Time to Open/Close	0

Table 3. Storage Node Characteristics

Storage Node	
Name	ST1
Inflows	No
Treatment	No
Invert elevation	580
Max depth	4
Initial depth	0

Ponded area	0
Storage Curve	Tabular
Curve Name	StorageCurve1

Table 4. Outfall Characteristics

Outfall 1	
Name	Out1
Inflows	No
Treatment	No
Invert elevation	583
Route to	Infil
Type	Free
Outfall 2	
Name	Out2
Inflows	No
Treatment	No
Invert elevation	580
Type	Free

Table 5. Storage Curve Characteristics

Storage Curve	
Name	StorageCurve1
Depth	Area
0	2400
1	2400
2	2400
3	2400
4	2400

Table 6. Aquifer Properties

Attribute	Value
Aquifer name	Aquifer1
Porosity	0.3
Wilting point	0.08
Field capacity	0.18

Attribute	Value
Conductivity	0.5
Conductivity slope	1
Tension slope	15.0
Upper evaporation fraction	0.35
Lower evaporation depth	14.0
Lower groundwater loss rate	0.01
Bottom elevation	189.37
Water table elevation	195.43
Unsaturated zone moisture	0.3
Upper evaporation pattern	

Table 7. Aquifer flow properties in Subcatchments

Attribute	Value
Subcatchment	1
Aquifer name	Aquifer1
Receiving node	Out2
Surface elevation	204.30
A1 coefficient	2
B1 exponent	1
A2 coefficient	0
B2 exponent	0
A3 coefficient	0
Surface water height	0
Channel bottom height	
Bottom elevation	
Water table elevation	
Unsaturated zone moisture	
Custom lateral flow equation	No
Custom deep flow equation	No

